

CONEPRO · REQUIREMENTS

ConePro requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/conepro		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.6	2026-09-30	Amish Chadha	Status from CNP-CAL-001 v0.4 after the design for construction (CNP-DDR-004); R5 stroke 939 mm; R2 lean term 1.8 mm; R13 \$335; R15 sensor set now includes the reel cable (line 17); no status changes
0.5	2026-09-27	Amish Chadha	Status from CNP-CAL-001 v0.3 after the reel housing change in CNP-DDR-003 (decided by Amish on 2026-09-27); R5 stroke 928 mm, 850 mm usable still met; R2 lean term 2.0 mm
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R5 restated to 850 mm per rod with the extension rod as a separate accessory; R10 met by the lighter plate, clamp and bag; status from CNP-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 status from CNP-CAL-001; decisions in CNP-DDR-001 (R6 met by D1, R9 assumes the lever of D2, new

R15 from D3); R10 total defined as
carried in the bag

0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

ConePro requirements

These requirements were checked by calculation at TRL 3 in CNP-CAL-001 v0.4 against the design decided in CNP-DDR-001, CNP-DDR-002 and CNP-DDR-003 and made constructable in CNP-DDR-004. None is now not met: two are at risk, nine are met on paper and four cannot be verified until hardware exists. The two at risk are **R2** (depth accuracy) and **R11** (shock survival). Version 0.4 applies Amish's 2026-09-25 acceptance of the review recommendations (CNP-DDR-002): **R5** is restated as 850 mm per lower rod, with a 500 mm extension rod offered as a separately carried accessory instead of a 1,000 mm clause (D10), and **R10** is now met at about 15.7 kg (was 16.7 kg) with a 6 mm plate, an aluminum clamp and arm and a lighter bag (D9). Targets are still proposals, not yet validated with users, and will be revised after co-design (see CNP-PRB-001). Version 0.5 reflects the larger reel housing decided by Amish on 2026-09-27 (CNP-DDR-003): R5 stays met with a 928 mm stroke, and the lean term in R2 grows from 1.8 mm to 2.0 mm, so R2 stays at risk. R15 carries decision D3.

Reference test. A single test point driven to 850 mm in a medium-stiff soil averaging 15 mm per blow (about 57 blows), by one person, on level ground.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status at TRL 3 (CNP-CAL-001)	Verification (later)
R1	Standard DCP geometry	8.0 kg hammer, 575 mm free drop, 16 mm rod, 20 mm cone with a 60 degree point, replaceable cone; tolerances as ASTM D6951	Met (nominal): model gives 8.00 kg and 575 mm; tolerances to check against the standard	Drawing check against the standard
R2	Depth measurement	Resolution 0.5 mm or finer; error within ± 1 mm over a 1,000 mm range compared with a steel rule	At risk: 0.046 mm resolution; bench error ± 0.98 mm worst case, ± 0.49 mm root sum square; a 2 degree lean adds up to 1.8 mm at 850 mm after tilt correction; usable range set at 850 mm (D11)	Bench calibration
R3	Blow detection	Every blow counted and no false counts from handling, over a 200-blow sequence	Not verifiable at TRL 3: Hall margin 2.3 times at the worst angle; three-signal logic (Hall, accelerometer peak, depth)	Bench blow sequence
R4	Per-blow record	Depth after each blow logged within 0.5 s of impact, with blow number and time	Met (calculation): settled within 35 ms, recorded at 0.21 s	Firmware sketch review
R5	Penetration range	850 mm or more per lower rod; deeper tests with an optional 500 mm extension rod carried as a separate accessory, with automatic re-zero (restated by D10; was 1,000 mm or more with one extension rod)	Met: 939 mm stroke, 850 mm usable per rod; extension rod in BOM line 16 (optional); re-zero not verifiable at TRL 3	Model check; field test
R6	Verticality	Warn the operator when the rod is more than 5 degrees off vertical	Met by design (D1): accelerometer reads tilt to about ± 0.3 degree; bubble level as backup	Bench tilt check
R7	Phone output	Live plot of depth against blows; DCP index per layer; indicative CBR from the ASTM D6951 correlations with the soil type selected; CSV export with time, location and operator notes	Not verifiable at TRL 3 (app specified, not written)	App specification review
R8	Works offline	No cell signal needed; logger stores 50 or more tests if the phone is absent or disconnects	Met (calculation): about 830 tests in 4 MB	Firmware sketch review
R9	One-person test time	Reference test, including setup and rod extraction with the optional lever (D2), in 12 min or less by one person	Met (estimate), thin: 8.1 to 11.7 min	Timed trials
R10	Portability	Total mass as carried in the bag, cells included and the optional lever and extension rod excluded, 16 kg or less; heaviest piece 10 kg or less; packed length 1.1 m or less	Met, thin: 15.7 kg (was 16.7 kg before D9); heaviest piece 9.8 kg; packed length about 1.08 m	Mass roll-up; weighing
R11	Rugged electronics		At risk: about 1,400 to 5,500 g mean at the anvil; about 540 g on the isolated	Drop-count test

ID	Requirement	Target	Status at TRL 3 (CNP-CAL-001)	Verification (later)
		Logger, sensors and connectors IP65; operate at 0 to 45 °C; anvil-mounted parts survive 10,000 blows	pad against 10,000 g part ratings; cable and connector fatigue not calculated	
R12	Battery life	8 h or more of continuous logging on user-replaceable non-lithium cells	Met (calculation): about 40 h, about 20 h at 0 °C	Power measurement
R13	Cost	Within the \$400 value-engineering target (hypothetical control target, not a limit) for one complete instrument in parts (optional lever and extension rod excluded)	Within the value-engineering target: \$335, \$65 under; \$401 with the lever and extension rod, \$1 over the target	Priced BOM (bom/ bom.csv)
R14	Honest results	App states the correlation used and that results are indicative and not for foundation design	Not verifiable at TRL 3 (app not written)	App specification review
R15	Retrofit fit (D3)	Sensor set (BOM items 8 to 12 and 17) fits a standard ASTM D6951 penetrometer with a 16 mm rod and a 50 to 80 mm anvil, without machining	Not verifiable at TRL 3: interfaces sized in the model; needs a commercial DCP to check	Fit check on a commercial DCP

■ Assumptions

- Soil behavior, blow rate and extraction time are estimates, not measurements. A practiced operator drops the hammer about once every 2.5 s; a slow one every 3.5 s.
- CBR correlations are those given in ASTM D6951 (from US Army Corps of Engineers work); other soils and regions may need local correlations.
- Mass estimates use steel at 7,850 kg/m³ and aluminum at 2,700 kg/m³ applied to the parametric model `cad/src/model.py`, plus catalog masses for bought parts (CNP-CAL-001, Table 2).
- Battery life assumes an average electronics draw of about 50 mA and about 2,000 mAh usable from AA alkaline cells at that current, halved at 0 °C.
- All other calculation assumptions are listed in CNP-CAL-001, Table 1.